THE ARTIN-TATE LEMMA: A SUBALGEBRA OVER WHICH A FINITELY GENERATED ALGEBRA IS FINITE IS OF FINITE TYPE

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Allowed standard prerequisites (editorial).

- Finitely generated algebras over a Noetherian ring are Noetherian. Used in the counted proof at lines 12527-12528. Makes the auxiliary subalgebra T' Noetherian, which is what allows the finiteness conclusion.
- Finiteness of a module detected after a surjection. Used at the end of the counted proof (lines 12536-12538). Transfers finiteness from a finite quotient back to the subalgebra T.

Lemma 0.1 (Artin-Tate)**.** *Let R be a Noetherian ring. Let S be a finitely generated R-algebra. If $T \subset S$ is an R-subalgebra such that S is finitely generated as a T-module, then T is of finite type over R.*

Proof. Choose elements $x_1, \ldots, x_n \in S$ which generate S as an R-algebra. Choose $y_1, \ldots, y_m$ in S which generate S as a T-module. Thus there exist $a_{ij} \in T$ such that $x_i = \sum a_{ij} y_j$. There also exist $b_{ijk} \in T$ such that $y_i y_j = \sum b_{ijk} y_k$. Let $T' \subset T$ be the sub R-algebra generated by a_{ij} and b_{ijk}. This is a finitely generated R-algebra, hence Noetherian. Consider the algebra

$$S' = T'[Y_1, \ldots, Y_m]/(Y_i Y_j - \sum b_{ijk} Y_k).$$

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Note that S' is finite over T', namely as a T'-module it is generated by the classes of $1, Y_1, \ldots, Y_m$. Consider the T'-algebra homomorphism $S' \to S$ which maps Y_i to y_i. Because $a_{ij} \in T'$ we see that x_j is in the image of this map. Thus $S' \to S$ is surjective. Therefore S is finite over T' as well. Since T' is Noetherian we conclude that $T \subset S$ is finite over T' and we win. $\square$